



Software Requirements Specification

For
CounselorPro

Version 1.1 (Revised)

23 August 2025

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1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for the CounselorPro application. It serves as a foundational agreement between the development team, stakeholders, and users.

1.2 Document Conventions

This document follows the IEEE 830-1998 standard. All requirements are uniquely identified for traceability. British International English spelling and grammar are used throughout.

1.3 Intended Audience and Reading Suggestions

This document is intended for project managers, software developers, the quality assurance team, and project stakeholders.

1.4 Product Scope

Product Scope

CounselorPro is an intelligent, web-based application designed to provide automated, personalised academic and extracurricular guidance by analysing scanned school term reports. The primary goal is to make high-quality, data-driven academic counselling accessible.

Version 1.0 (MVP) Scope: The initial release will focus on perfecting the core data pipeline. The scope includes: the user account system, the file upload mechanism, the OCR-based data extraction and user-verification workflow, and a **rules-based recommendation engine**. The rules-based engine will provide foundational, template-driven advice based on the verified data. More advanced, generative AI features are planned for future versions.

1.5 References

- IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications*.
- AUCDT Brand Identity Guide v2.0, August 2025.

2. Overall Description

2.1 Product Perspective

CounselorPro is a new, self-contained, standalone web application. Future versions may include APIs for integration with School Information Systems (SIS).

2.2 Product Functions

The major functions are: User Account Management, Report Upload, Data Extraction & Verification, Performance Analysis, Recommendation Generation, and Report Generation.

2.3 User Classes and Characteristics

User Classes

2.4 Operating Environment

The application will be a web-based platform accessible through modern web browsers on desktop and mobile devices, hosted on a secure cloud infrastructure.

2.5 Design and Implementation Constraints

i Important Note

The system must comply with data protection regulations such as GDPR and FERPA. This is a critical, non-negotiable constraint requiring dedicated resources for security architecture, data handling protocols, and consent management.

- ⚠ C-1 (Data Privacy):** Must comply with GDPR/FERPA.
- ⚠ C-2 (OCR):** Must support multiple languages, starting with English.
- ⚠ C-3 (AI Ethics):** The recommendation engine must avoid generating biased or deterministic advice. For V1.0, this will be managed via a curated, rules-based system.
- ⚠ C-4 (Branding):** The UI must strictly adhere to the AUCDT Brand Identity Guide v2.0.

2.6 Assumptions and Dependencies

🍪 **Colour Palette & Typography:**

🍪 **Containers & Buttons:**

ID

FR-1: User Account Management

FR-1.1

FR-2: Report Upload

FR-2.1

FR-3: Data Extraction

FR-3.1

FR-3.2

FR-4: Performance Analysis

FR-4.1

FR-4.2

FR-5: Recommendation Generation

FR-5.1 **rules-based engine**

FR-5.2 **Acceptance Criterion:**

FR-6: Report Generation & Display

FR-6.1

NFR-1.1:

NFR-1.2:

NFR-2.1:

NFR-2.2:

★ **NFR-3.1 (Reliability):**

★ **NFR-3.2 (Usability):**

✔ **NFR-4.1 (OCR Accuracy):**

❗ **EH-1:**

❗ **EH-2:**

❗ **EH-3:**